

CSE340: Computer Architecture

Quiz 1

Time: 40 minutes

Semester: Summer 26

Total Marks: 22

Name: _____ ID: _____ Section: _____

Question 1

Qualcomm is a manufacturing company of Snapdragon processors. Their manufacturing process starts with a wafer of 300mm diameter with a cost of 120\$. After the manufacturing process ends, they have 415 working dies.

- a. If their yield is 89%, model parameter $N = 1.7$, find out the following values. Your answer must be in the question and all calculations shown in the answer script. [8]

Wafer Area	Die per wafer	Die Area	Cost per die
70686 mm ²	466	151.686 mm ²	0.289

Each Snapdragon processor was tested in a reference computer which ran a file compression program in 1372s. You bought one of the processors and to run the same program, your CPU took 520 clock cycles.

- b. The instruction count of the program is 350. As you know how to calculate SPECRatio, you found it to be 0.8. Calculate the following values. Your answer must be in the question, calculations must be shown in the answer script. [8]

Execution Time	Average CPI	Clock Rate	Clock Period
1715s	1.486	0.303 hz	3.3s

- c. According to the SPECRatio you found in question b, is your processor working as it should? Explain why/why not. [2]

Answer:

Execution time greater than reference time.

Question 2

- a.** Explain what you understand by “Performance via prediction” in the seven great ideas with an example. [2]

Answer:

- b.** Explain how an algorithm affects performance with an example. [2]

Answer:

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Question 1

AMD is a manufacturing company of Ryzen processors. Their manufacturing process starts with a wafer of 175mm radius with a cost of 150\$. After the manufacturing process ended, their yield was 92%.

- a. If dies per wafer is 490, model parameter $N = 1.7$, find out the following values. Your answer must be in the question and all calculations shown in the answer script. [8]

Wafer Area	No of Working Dies	Die Area	Cost per die
96211.28 mm ²	450	196.34 mm ²	0.33

Each Ryzen processor was tested in a reference computer which ran a game in 1520s. You bought one of the processors and to run the same program, your CPU took 1605s and 960 clock cycles.

- b. The average CPI of the program is 4. Calculate the following values. Your answer must be in the question, calculations must be shown in the answer script. [8]

SPECRatio	Instruction Count	Clock Rate	Clock Period
0.95	240	0.598 hz	1.695s

- c. According to the SPECRatio you found in question b, is your processor working as it should? Explain why/why not. [2]

Answer:

Execution time greater than reference time.

Question 2

- a.** Explain what you understand by “Hierarchy of memories” in the seven great ideas with an example. [2]

Answer:

- b.** Explain how an I/O system (Hardware and OS) affects performance with an example. [2]

Answer:

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Question 1

Apple is a manufacturing company of Apple M4 processors. The die area of an M4 chip is 79.8 mm^2 . They can make around 550 dies out of a wafer.

- a. If the number of working dies is 508 and each die costs 7\$, model parameter $N = 2$, find out the following values. Your answer must be in the question and all calculations shown in the answer script. [8]

Wafer Area	Yield	Defects/ mm^2	Cost per wafer
43934 mm^2	92%	0.0053	3556

Each M4 processor was tested in a reference computer which ran a game in 1440s. They wanted to improve the overall performance 5 times. A specific operation of the game took 1200s.

- b. The average CPI of the program is 4 and the clock period is 2s. Calculate the following values. Your answer must be in the question, calculations must be shown in the answer script. [8]

T_improved	Improvement factor	Instruction count	Performance after improvement
288s	25	180	0.0034

- c. Explain which seven great ideas the manufacturer used to increase performance and why. [2]

Answer:

Common case fast for maximum effect.

Question 2

- a. Explain what you understand by “Performance via parallelism” in the seven great ideas with an example. [2]

Answer:

- b. Explain how a Programming Language affects performance with an example. [2]

Answer:

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Question 1

MediaTek is a manufacturing company of Dimensity processors. The die area of a chip is 140.79 mm^2 and each die costs 1.4\$. They can make around 390 dies out of a wafer.

- a. If the number of working dies is 375, model parameter $N = 2$, find out the following values. Your answer must be in the question and all calculations shown in the answer script. [8]

Wafer Diameter	Yield	Defects/ mm^2	Cost per wafer
264.406 mm	96.15%	0.00014	525

Each Dimensity processor was tested in a reference computer which ran a video player in 950s . They wanted to improve the overall performance 3 times. A specific operation of the video player took 720s.

- b. The instruction count of the program is 320 and the clock period is 1.75s. Calculate the following values. Your answer must be in the question, calculations must be shown in the answer script. [8]

T_improved	Improvement factor	Average CPI	Performance before improvement
316.667s	8.307	1.696	0.001

- c. Explain which law the manufacturer used to increase performance and how much performance was increased. [2]

Answer:

Amdahl's law.

Performance improvement = 0.002

Question 2

- a. Explain what you understand by “Dependability via redundancy” in the seven great ideas with an example. [2]

Answer:

- b. Explain how a Processor and memory system affects performance with an example. [2]

Answer: